

Project Design Phase
Solution Architecture

Date	15 February 2025
Team ID	LTVIP2025TMID54870
Project Name	DocSpot: Seamless Appointment Booking for Health
Maximum Marks	4 Marks

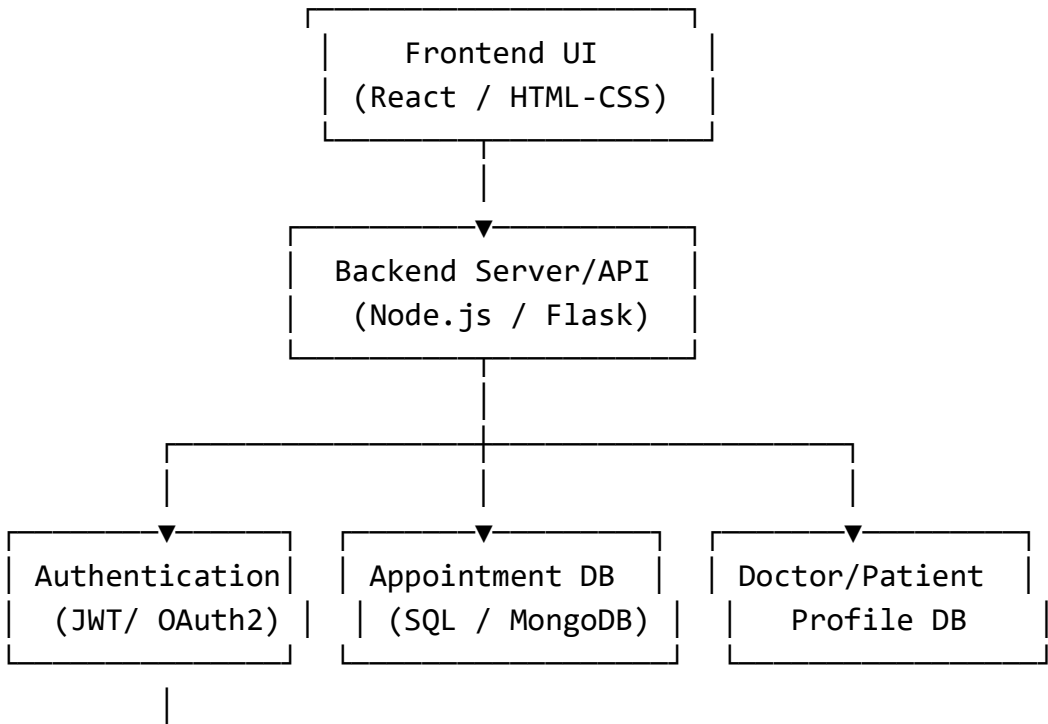
Solution Architecture:

 Solution Architecture Description:

Goal:
To design and implement a robust, scalable, and secure web-based application that allows users to search, book, and manage doctor appointments seamlessly, while offering doctors a dashboard to manage their availability, appointments, and patient feedback.

 Solution Architecture Diagram (Textual Representation):

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Notification

 (Email/SMS Reminders)

- Optional Add-ons (Scalable Modules):
- Telemedicine Integration (Jitsi/WebRTC)
 - Admin Dashboard & Analytics
 - Payment Gateway (Razorpay/Stripe)

 Technologies & Tools Used:

- **Frontend:** React.js, Tailwind CSS (or Bootstrap)
- **Backend:** Node.js with Express or Python Flask
- **Database:** MongoDB (NoSQL) or PostgreSQL (SQL)
- **Authentication:** JWT-based secured login (OAuth 2.0 optional)
- **Hosting:** AWS EC2 / Render / Heroku
- **Notifications:** Twilio (SMS) or SendGrid (Email)
- **APIs:** REST APIs for communication between frontend and backend

 Features in Phases:

Phase	Features
Phase 1	Basic doctor search, booking, login/logout
Phase 2	Doctor-side dashboard, appointment status, rescheduling
Phase 3	Feedback module, notification system, data analytics
Phase 4	Teleconsultation, admin dashboard, multi-language support

 Security Considerations:

- Encrypted JWT tokens for secure sessions
- HTTPS enforced for all communications
- Rate limiting & input sanitization to prevent abuse

